

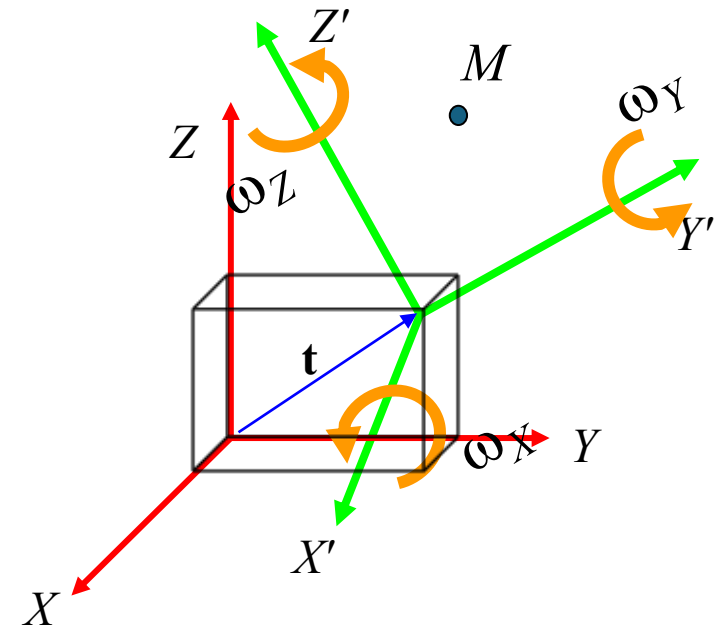


Transformaciones 3D

(CAPÍTULO II – CV02C)

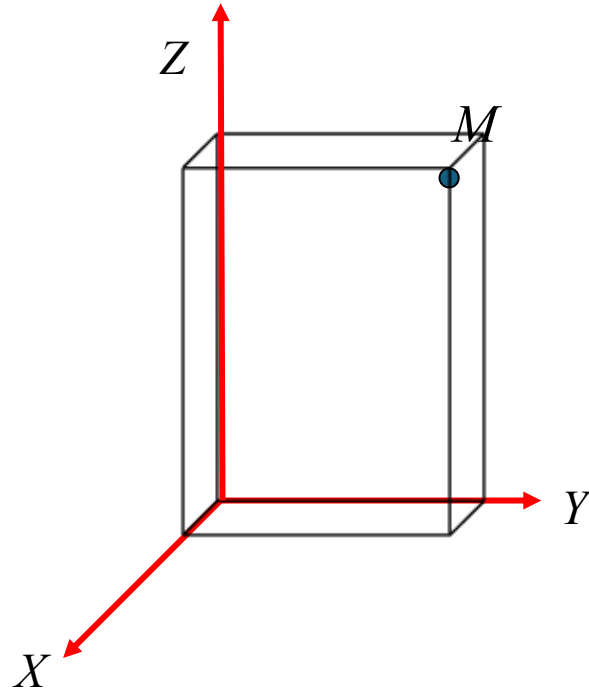
Domingo Mery 

Ciencia de la Computación
Universidad Católica – Chile
<http://domingomery.ing.uc.cl>



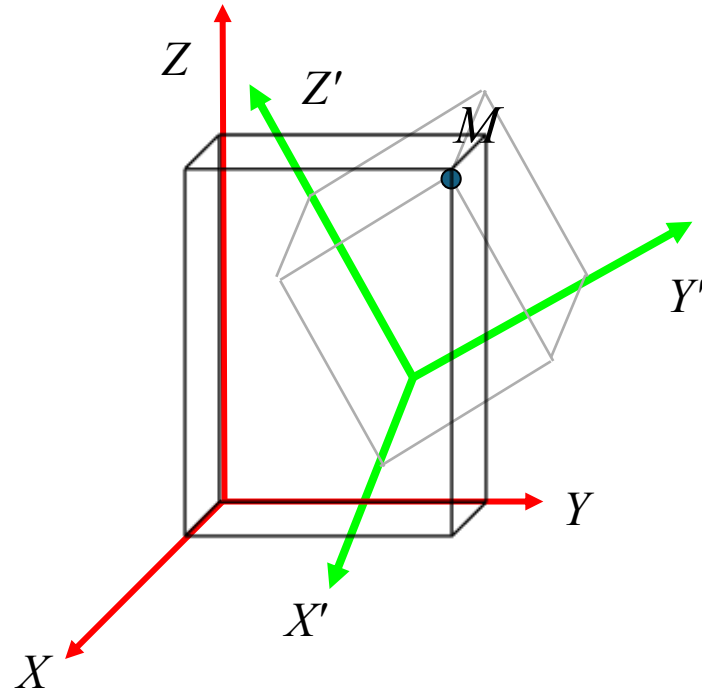
Transformaciones 3D

[3D Transformation]



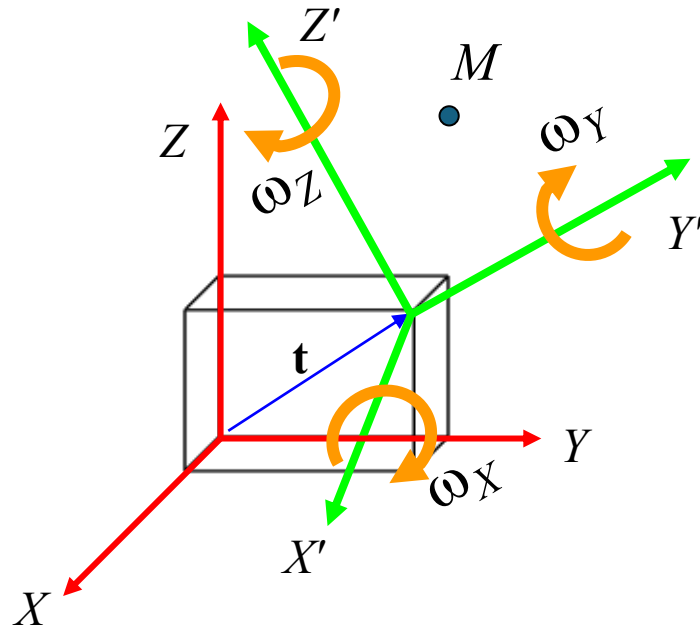
$$\mathbf{M} = [X \ Y \ Z \ 1]^T$$

[3D Transformation]



$$\mathbf{M} = [X \ Y \ Z \ 1]^T$$

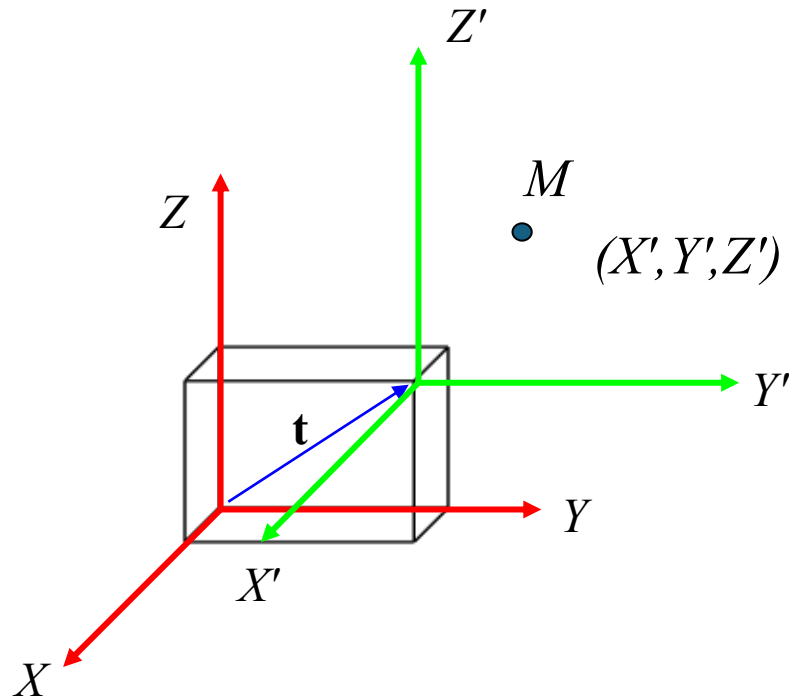
$$\mathbf{M}' = [X' \ Y' \ Z' \ 1]^T$$



$\mathbf{t}=(t_X, t_Y, t_Z)$: Translation
 $\omega_X, \omega_Y, \omega_Z$: Rotation

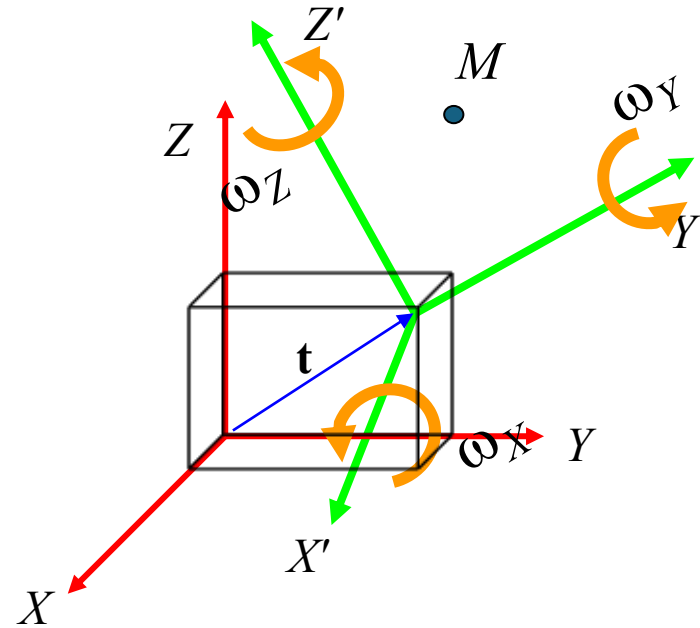
[3D Transformation]

3D Translation:

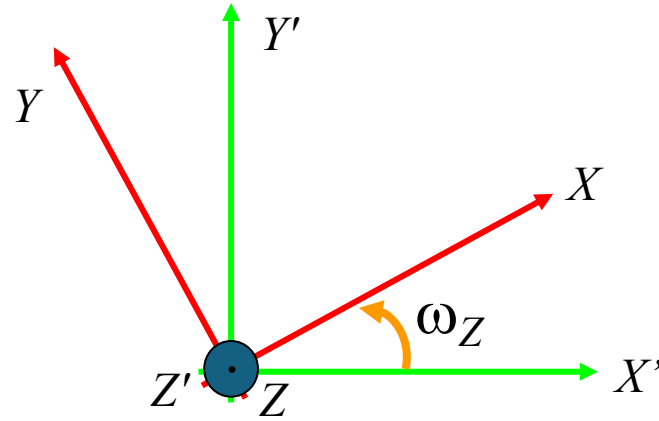


$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} X' + t_x \\ Y' + t_y \\ Z' + t_z \end{bmatrix} \quad \longrightarrow \quad \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

3D Rotation:

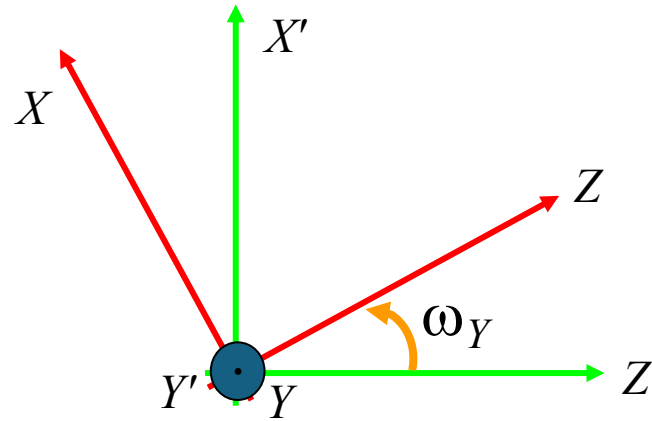


Axis Z Rotation:



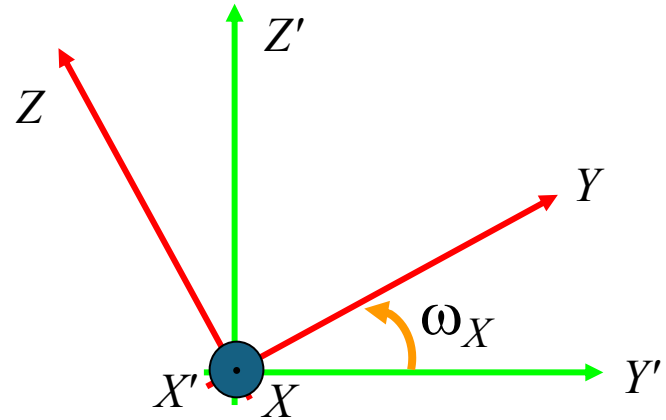
$$\mathbf{R}_Z = \begin{bmatrix} \cos(\omega_Z) & \sin(\omega_Z) & 0 \\ -\sin(\omega_Z) & \cos(\omega_Z) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Axis Y Rotation:



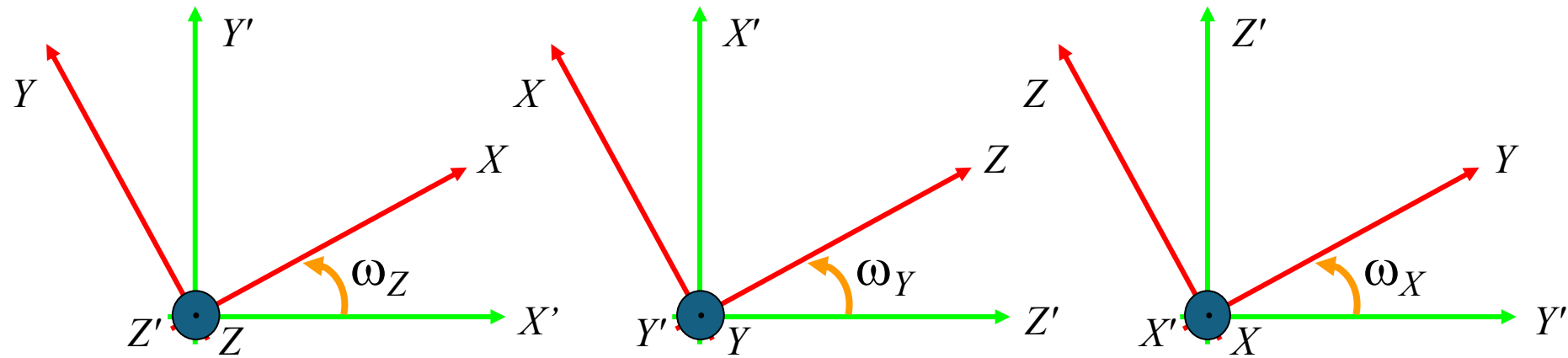
$$\mathbf{R}_Y = \begin{bmatrix} \cos(\omega_Y) & 0 & -\sin(\omega_Y) \\ 0 & 1 & 0 \\ \sin(\omega_Y) & 0 & \cos(\omega_Y) \end{bmatrix}$$

Axis X Rotation :



$$\mathbf{R}_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\omega_X) & \sin(\omega_X) \\ 0 & -\sin(\omega_X) & \cos(\omega_X) \end{bmatrix}$$

[3D Transformation]



$$\mathbf{R}_Z = \begin{bmatrix} \cos(\omega_Z) & \sin(\omega_Z) & 0 \\ -\sin(\omega_Z) & \cos(\omega_Z) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_Y = \begin{bmatrix} \cos(\omega_Y) & 0 & -\sin(\omega_Y) \\ 0 & 1 & 0 \\ \sin(\omega_Y) & 0 & \cos(\omega_Y) \end{bmatrix}$$

$$\mathbf{R}_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\omega_X) & \sin(\omega_X) \\ 0 & -\sin(\omega_X) & \cos(\omega_X) \end{bmatrix}$$

$$\mathbf{R}(\omega_X, \omega_Y, \omega_Z) = \mathbf{R}_X(\omega_X) \mathbf{R}_Y(\omega_Y) \mathbf{R}_Z(\omega_Z) = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

$$R_{11} = \cos(\omega_Y) \cos(\omega_Z)$$

$$R_{12} = \cos(\omega_Y) \sin(\omega_Z)$$

$$R_{13} = -\sin(\omega_Y)$$

$$R_{21} = \sin(\omega_X) \sin(\omega_Y) \cos(\omega_Z) - \cos(\omega_X) \sin(\omega_Z)$$

$$R_{22} = \sin(\omega_X) \sin(\omega_Y) \sin(\omega_Z) + \cos(\omega_X) \cos(\omega_Z)$$

$$R_{23} = \sin(\omega_X) \cos(\omega_Y)$$

$$R_{31} = \cos(\omega_X) \sin(\omega_Y) \cos(\omega_Z) + \sin(\omega_X) \sin(\omega_Z)$$

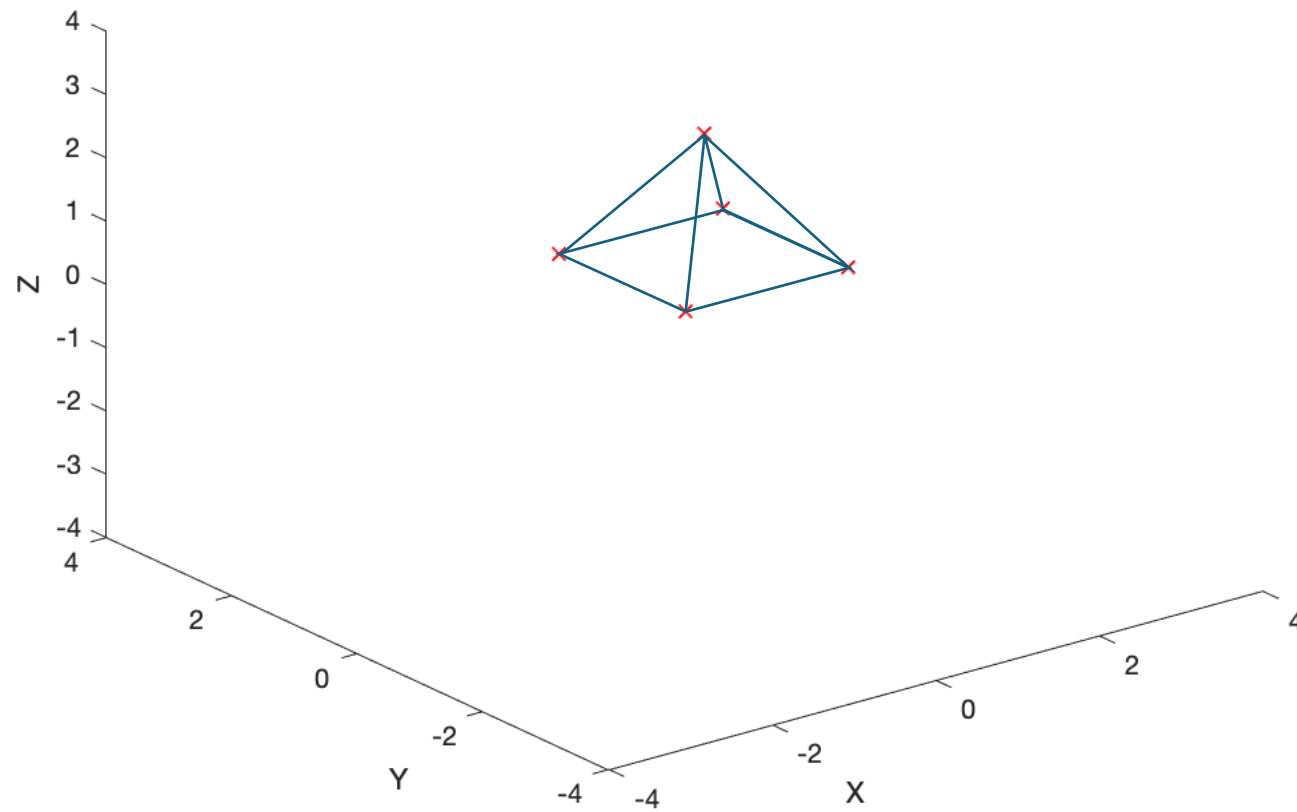
$$R_{32} = \cos(\omega_X) \sin(\omega_Y) \sin(\omega_Z) - \sin(\omega_X) \cos(\omega_Z)$$

$$R_{33} = \cos(\omega_X) \cos(\omega_Y)$$

[3D Transformation]

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

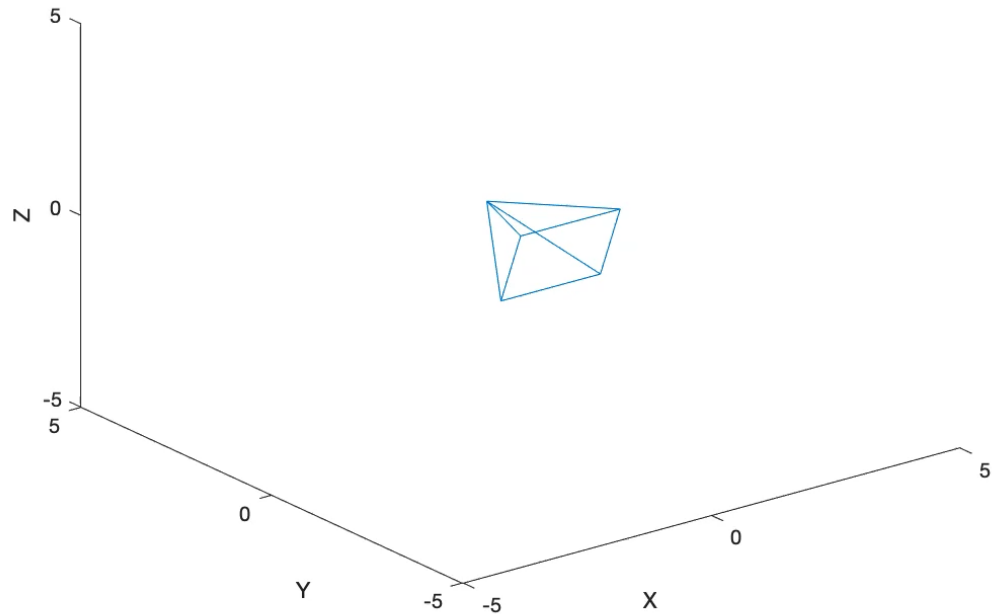
[Example: Pyramid]



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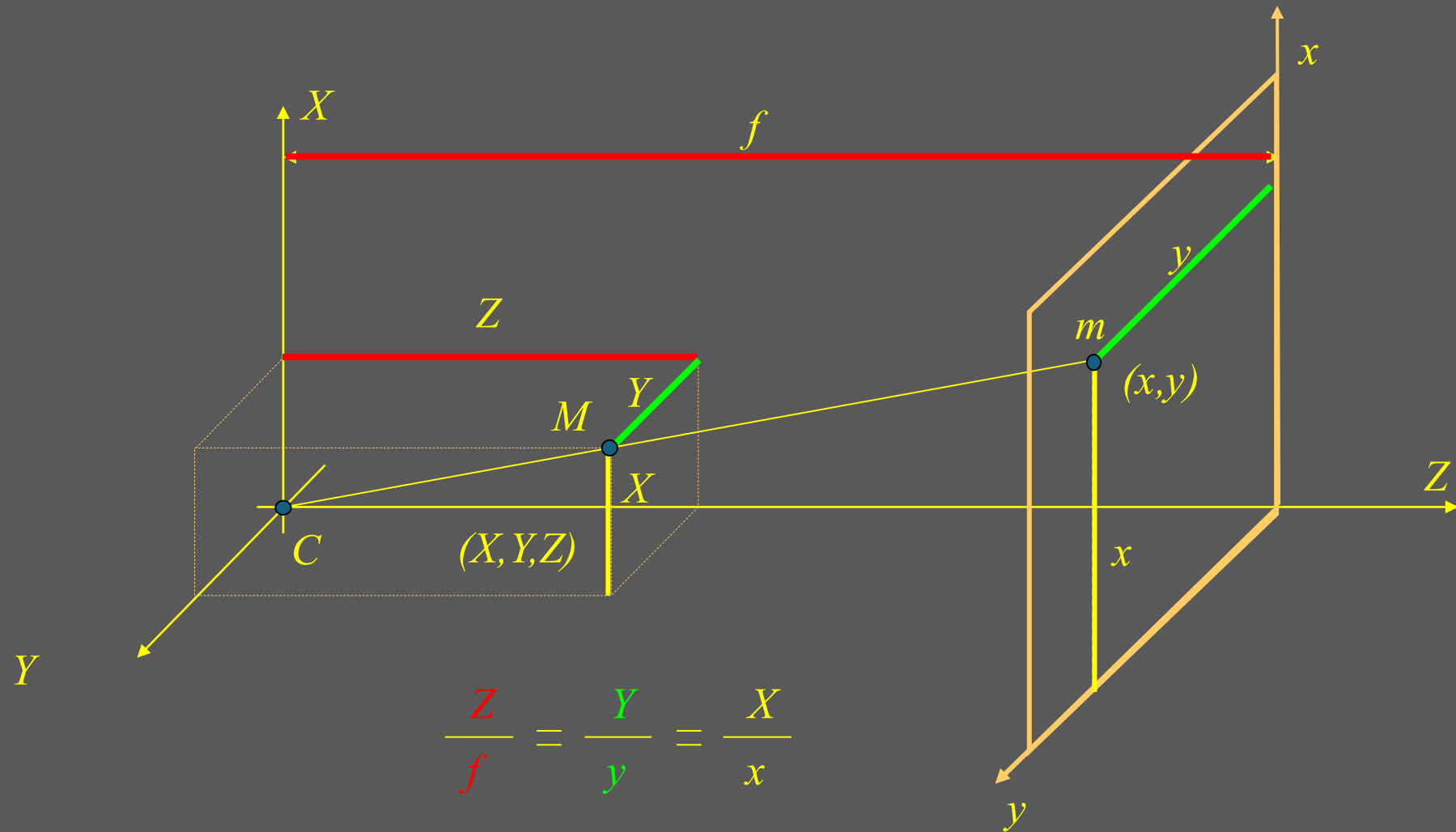
t = ( 0.00, 0.00, 0.00) w = ( 0.00, 0.00, 0.00)
t = ( 1.00, 2.00, 3.00) w = ( 0.00, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.57, 0.00, 0.00)
t = ( 0.00, 0.00, 2.00) w = ( 3.14, 0.00, 0.00)
t = ( 2.00, 2.00, 0.00) w = ( 0.00, 0.79, -0.39)
starting pyramid...
t = ( 0.00, 0.00, 0.00) w = ( 0.00, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.09, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.17, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.26, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.35, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.44, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.52, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.61, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.70, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.79, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.87, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.96, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.05, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.13, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.22, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.31, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.40, 0.00, 0.00)

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Transformaciones 3D $\rightarrow$ 2D

[Perspective Projection]



[Perspective Projection]

$$\frac{\overset{Z}{f}}{\underset{f}{f}} = \frac{\overset{Y}{y}}{\underset{y}{y}} = \frac{\overset{X}{x}}{\underset{x}{x}}$$

$Zx = f X$
 $Zy = f Y$

$$Z \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} fX \\ fY \\ Z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

m

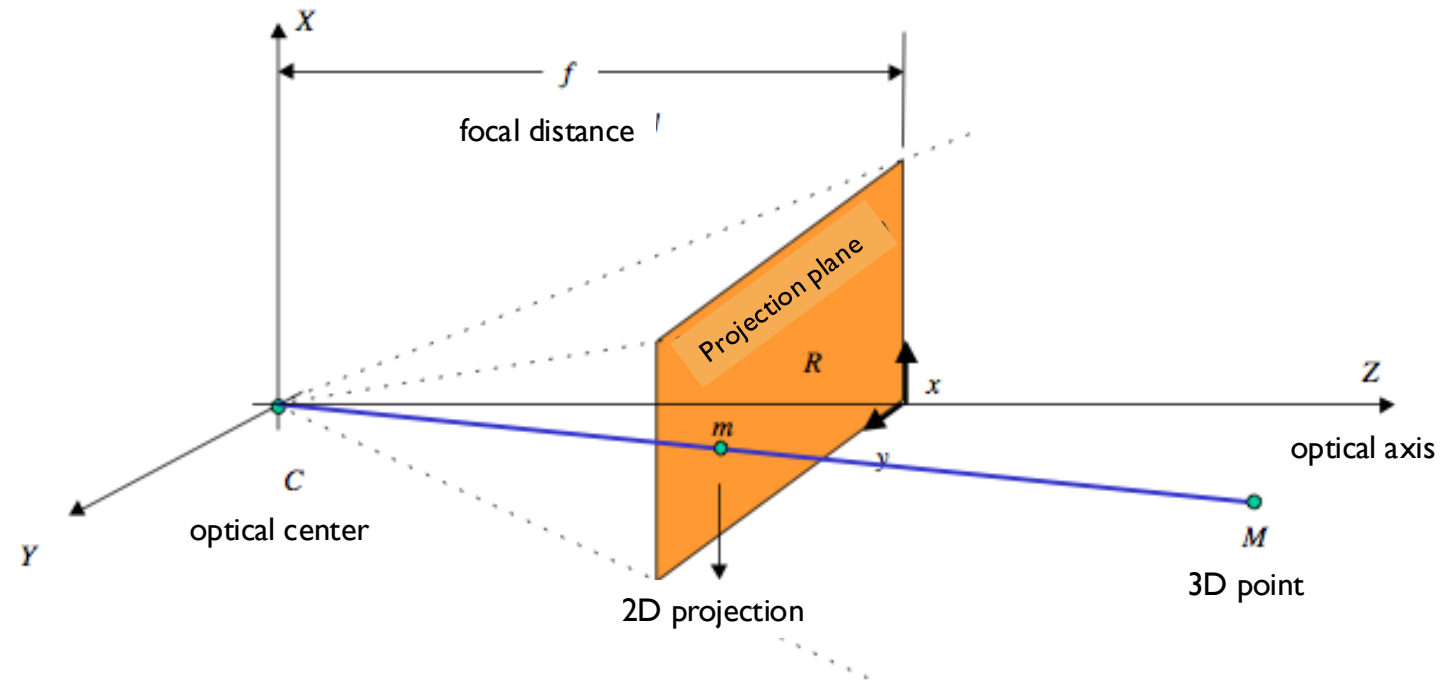
is proportional to

P

*

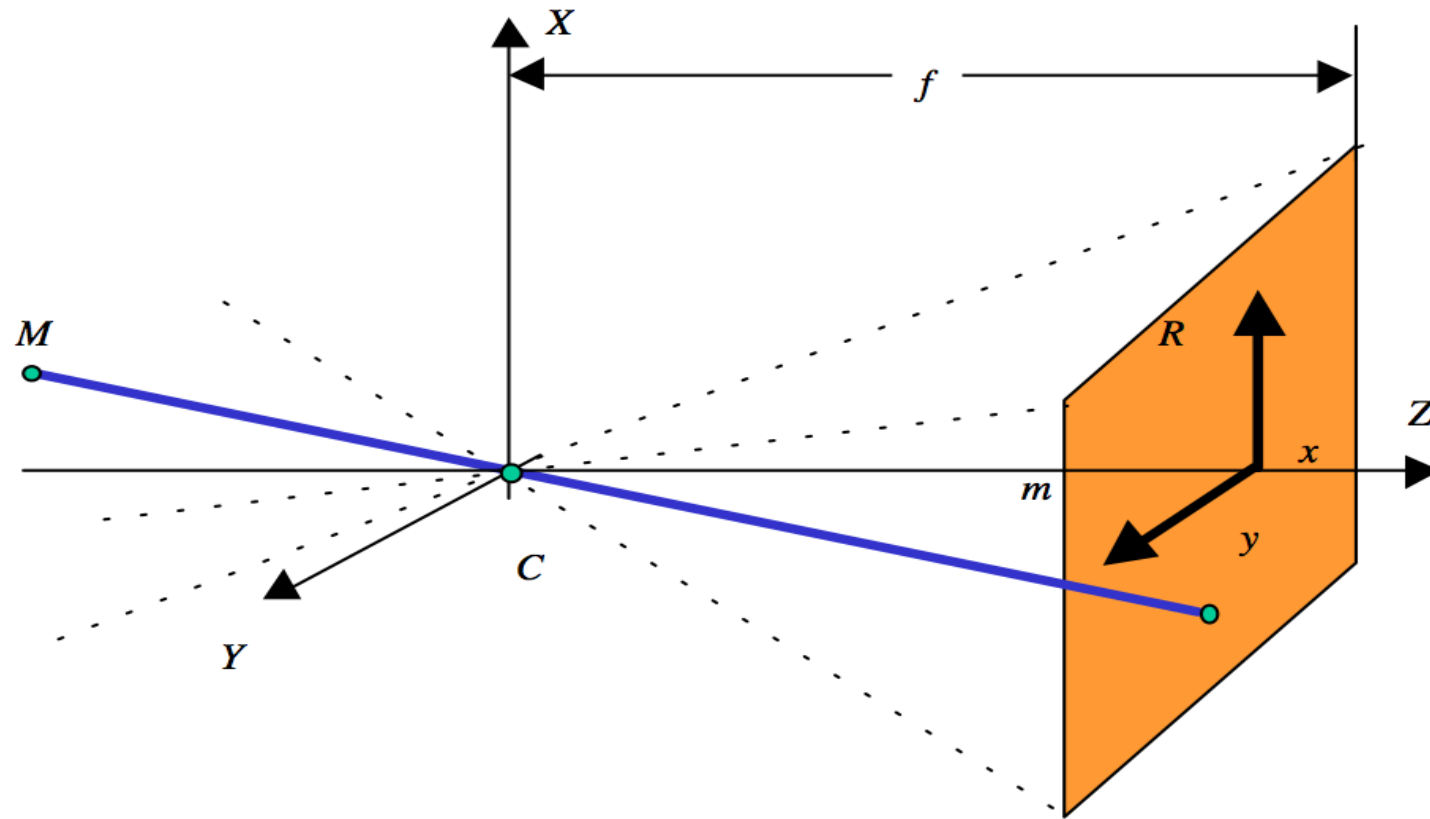
M

$$\lambda \mathbf{m} = \mathbf{P} \mathbf{M}$$



Matrix $P = ?$

$$\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

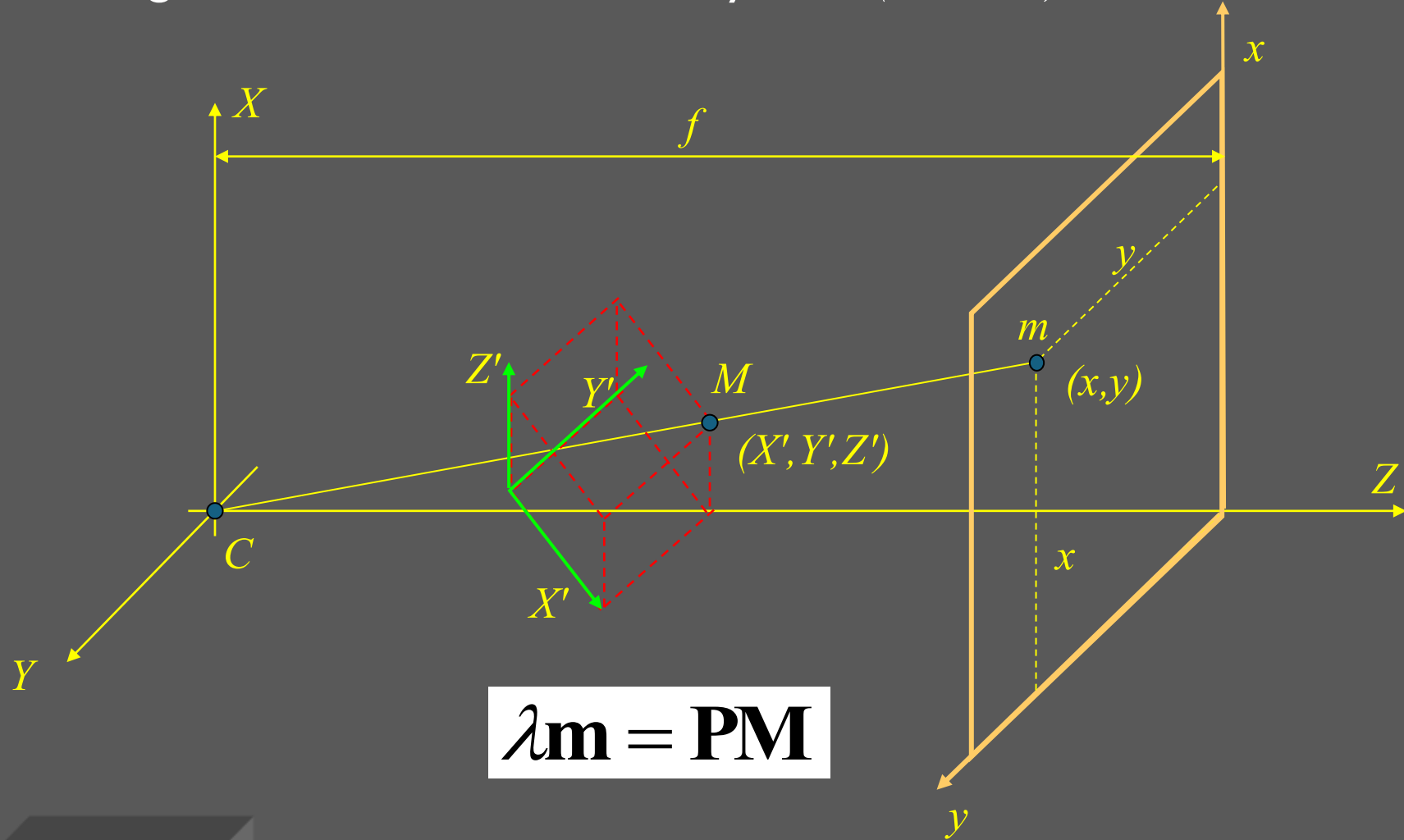


Matrix $P = ?$

$$\begin{bmatrix} -f & 0 & 0 & 0 \\ 0 & -f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

[Perspective Projection]

Now, M is given in a new coordinate system: (X', Y', Z')

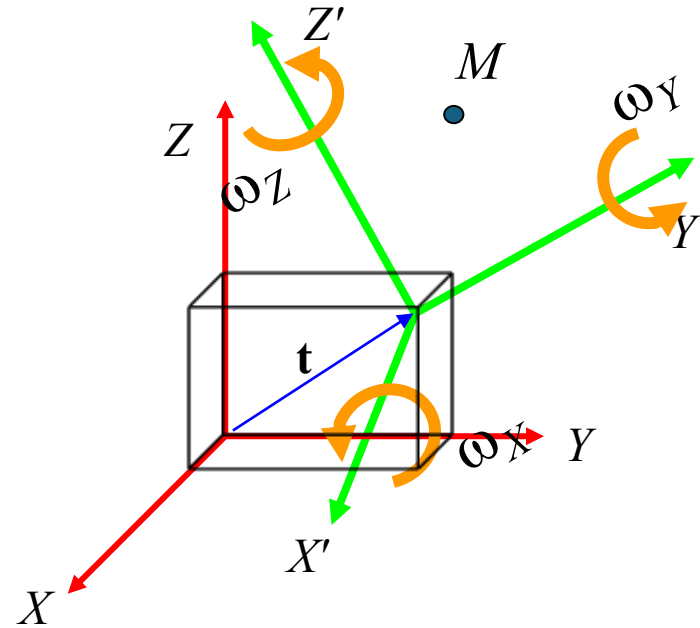


$$\lambda \mathbf{m} = \mathbf{P} \mathbf{M}$$

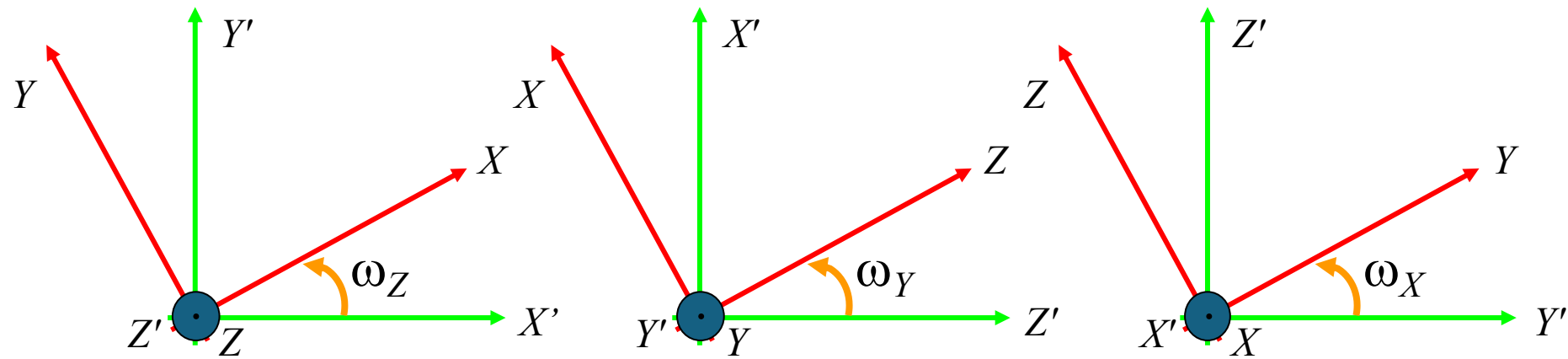
$$\lambda \mathbf{m} = \mathbf{P} \mathbf{M}$$

However, $\mathbf{M}$ was given in (X, Y, Z) coordinate system

3D Rotation:



[3D Transformation]



$$\mathbf{R}_Z = \begin{bmatrix} \cos(\omega_Z) & \sin(\omega_Z) & 0 \\ -\sin(\omega_Z) & \cos(\omega_Z) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_Y = \begin{bmatrix} \cos(\omega_Y) & 0 & -\sin(\omega_Y) \\ 0 & 1 & 0 \\ \sin(\omega_Y) & 0 & \cos(\omega_Y) \end{bmatrix}$$

$$\mathbf{R}_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\omega_X) & \sin(\omega_X) \\ 0 & -\sin(\omega_X) & \cos(\omega_X) \end{bmatrix}$$

$$\mathbf{R}(\omega_X, \omega_Y, \omega_Z) = \mathbf{R}_X(\omega_X) \mathbf{R}_Y(\omega_Y) \mathbf{R}_Z(\omega_Z) = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

$$R_{11} = \cos(\omega_Y) \cos(\omega_Z)$$

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$$R_{13} = -\sin(\omega_Y)$$

$$R_{21} = \sin(\omega_X) \sin(\omega_Y) \cos(\omega_Z) - \cos(\omega_X) \sin(\omega_Z)$$

$$R_{22} = \sin(\omega_X) \sin(\omega_Y) \sin(\omega_Z) + \cos(\omega_X) \cos(\omega_Z)$$

$$R_{23} = \sin(\omega_X) \cos(\omega_Y)$$

$$R_{31} = \cos(\omega_X) \sin(\omega_Y) \cos(\omega_Z) + \sin(\omega_X) \sin(\omega_Z)$$

$$R_{32} = \cos(\omega_X) \sin(\omega_Y) \sin(\omega_Z) - \sin(\omega_X) \cos(\omega_Z)$$

$$R_{33} = \cos(\omega_X) \cos(\omega_Y)$$

[3D Transformation]

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Perspective Projection]

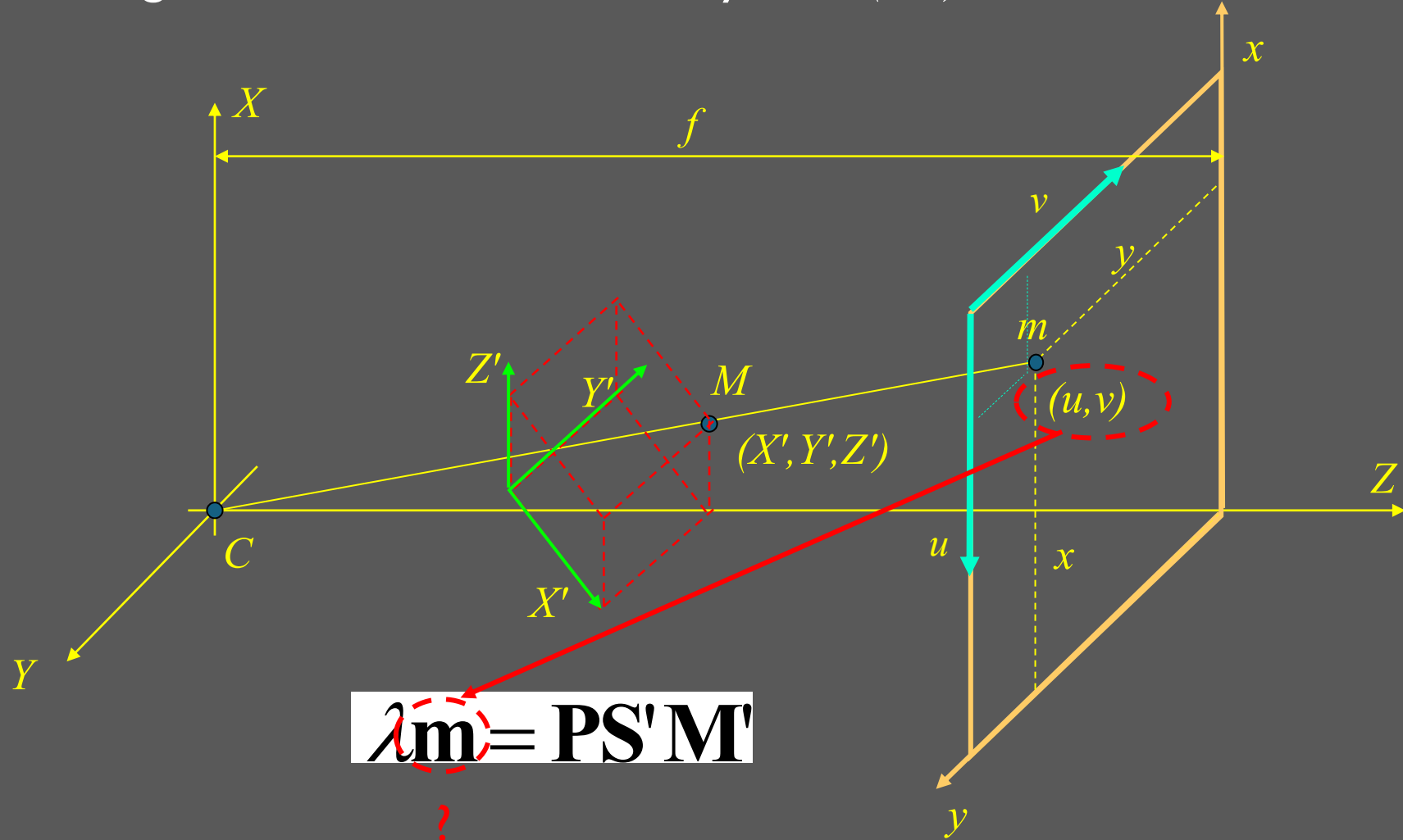
$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix} \longrightarrow \mathbf{M} = \underbrace{\mathbf{S}' \mathbf{M}'}_{\text{}}$$

$$\lambda \mathbf{m} = \mathbf{P} \mathbf{M}$$

$$\lambda \mathbf{m} = \underbrace{\mathbf{P} \mathbf{S}'}_{\text{}} \mathbf{M}'$$

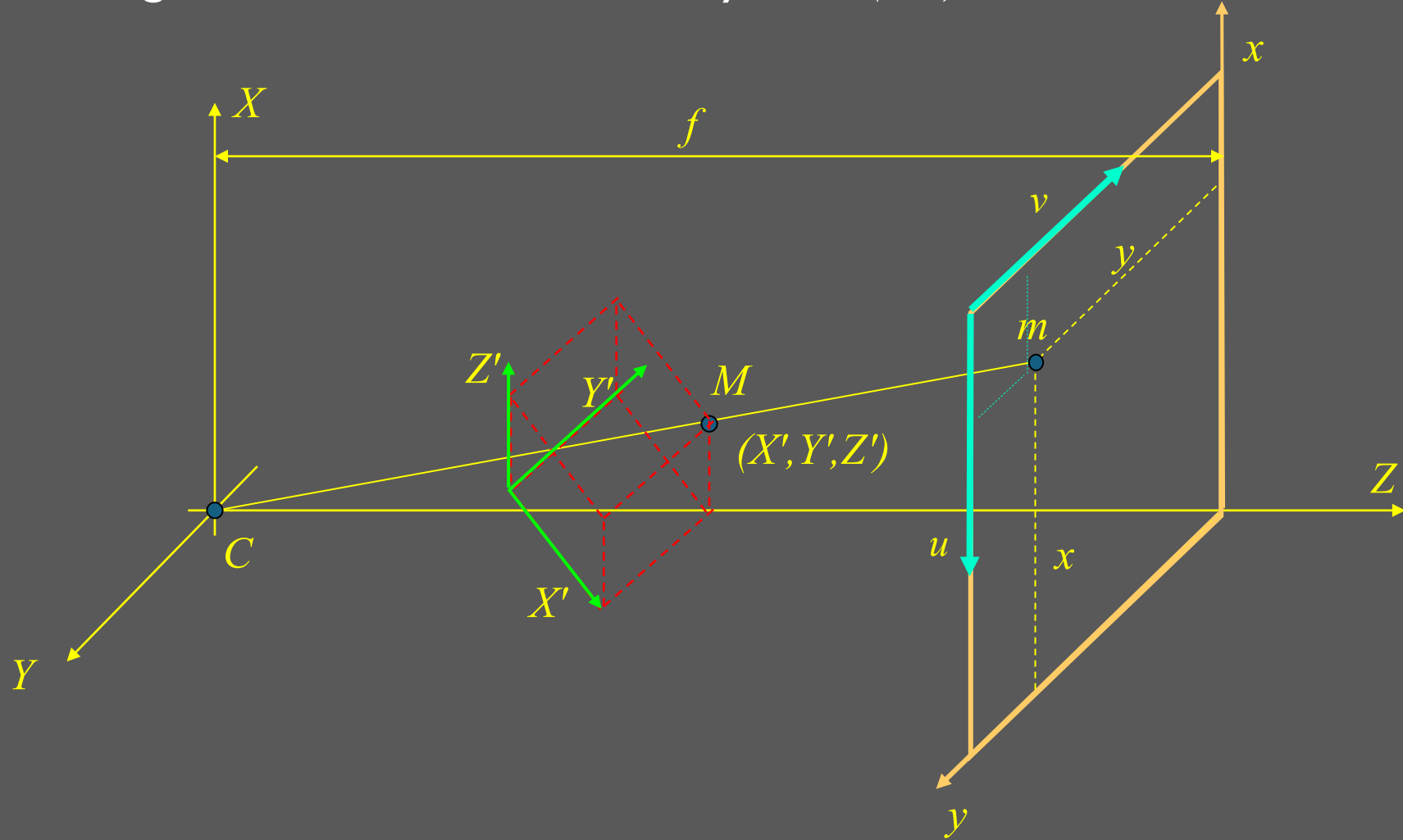
[Perspective Projection]

Now, m is given in a new coordinate system: (u, v)



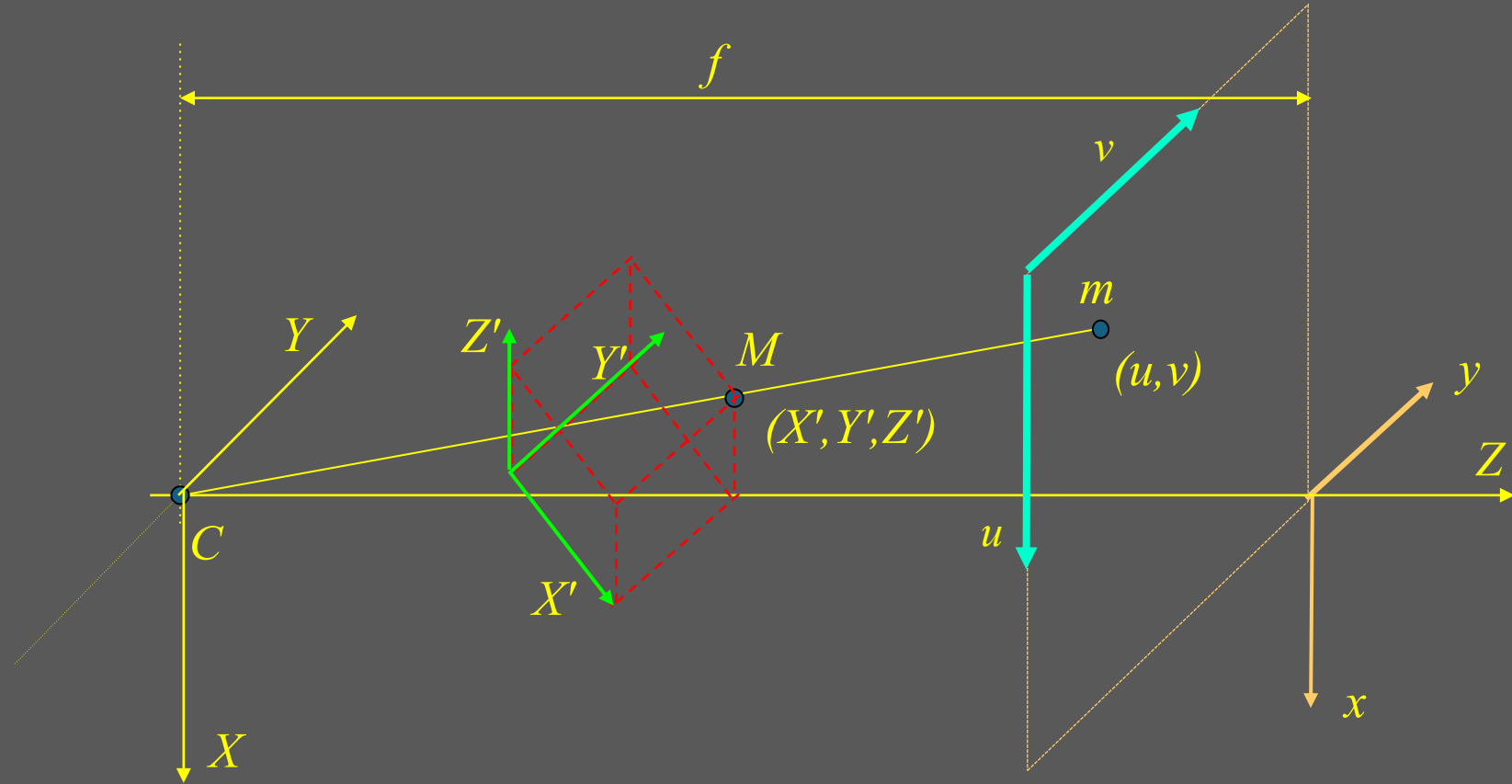
[Perspective Projection]

Now, m is given in a new coordinate system: (u, v)

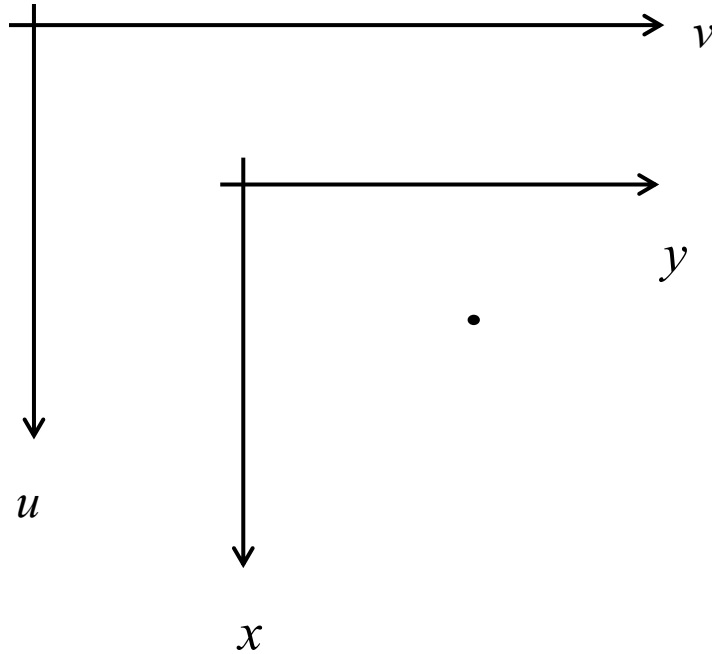


[Perspective Projection]

Now, m is given in a new coordinate system: (u, v)



[Perspective Projection]



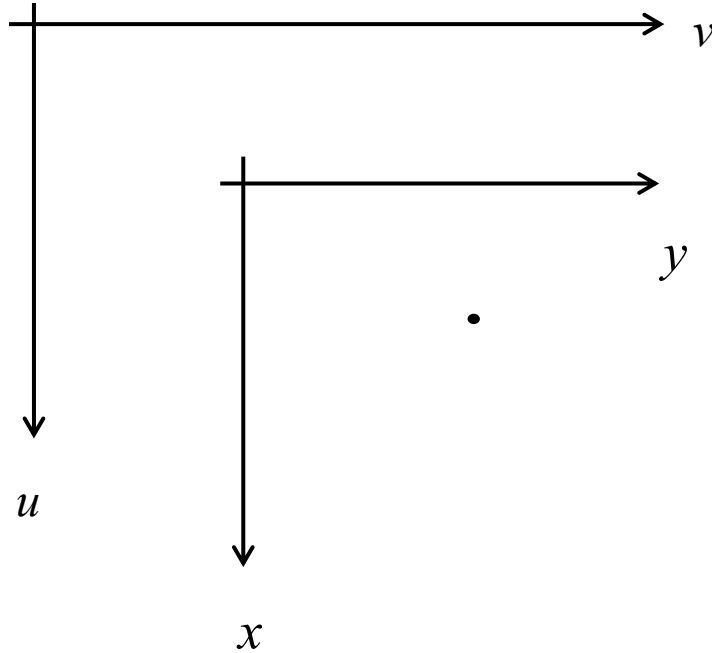
Between (x, y) and (u, v) there is no rotation.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & u_0 \\ 0 & 1 & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK?

No, because in a camera system (x, y) is given in mm or inches and (u, v) in pixels.

[Perspective Projection]



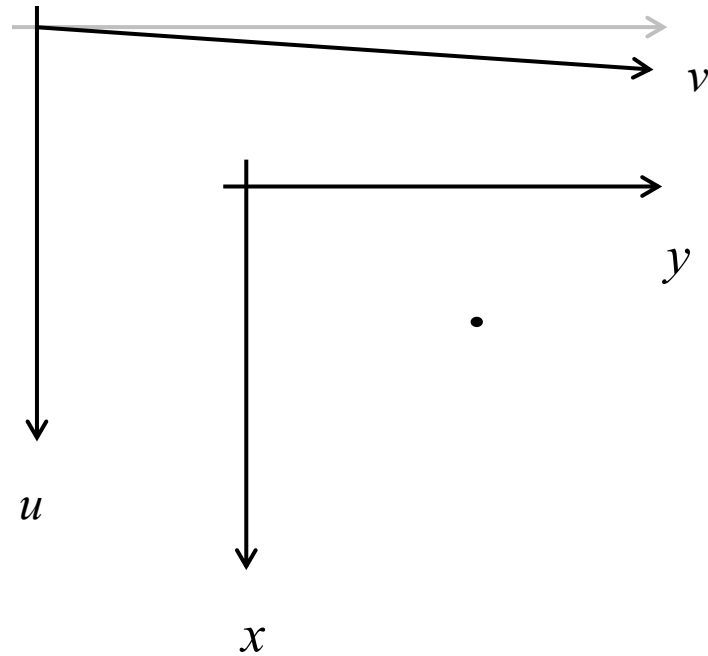
Between (x, y) and (u, v) there is no rotation.
There is a scale factor in each direction.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK?

No, because sometimes the angle between u and v is not 90, it could be for example 89.99.

[Perspective Projection]



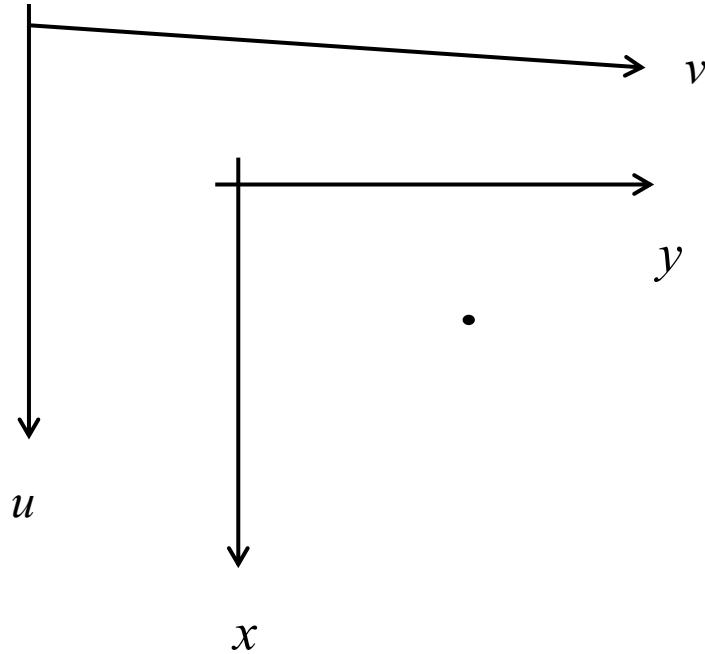
Between (x, y) and (u, v) there is no rotation.
There is a scale factor in each direction.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK?

No, because sometimes the angle between u and v is not 90, it could be for example 89.99.

[Perspective Projection]



Between (x, y) and (u, v) there is no rotation.
There is a scale factor in each direction.
There is a skew factor.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK? Yes!

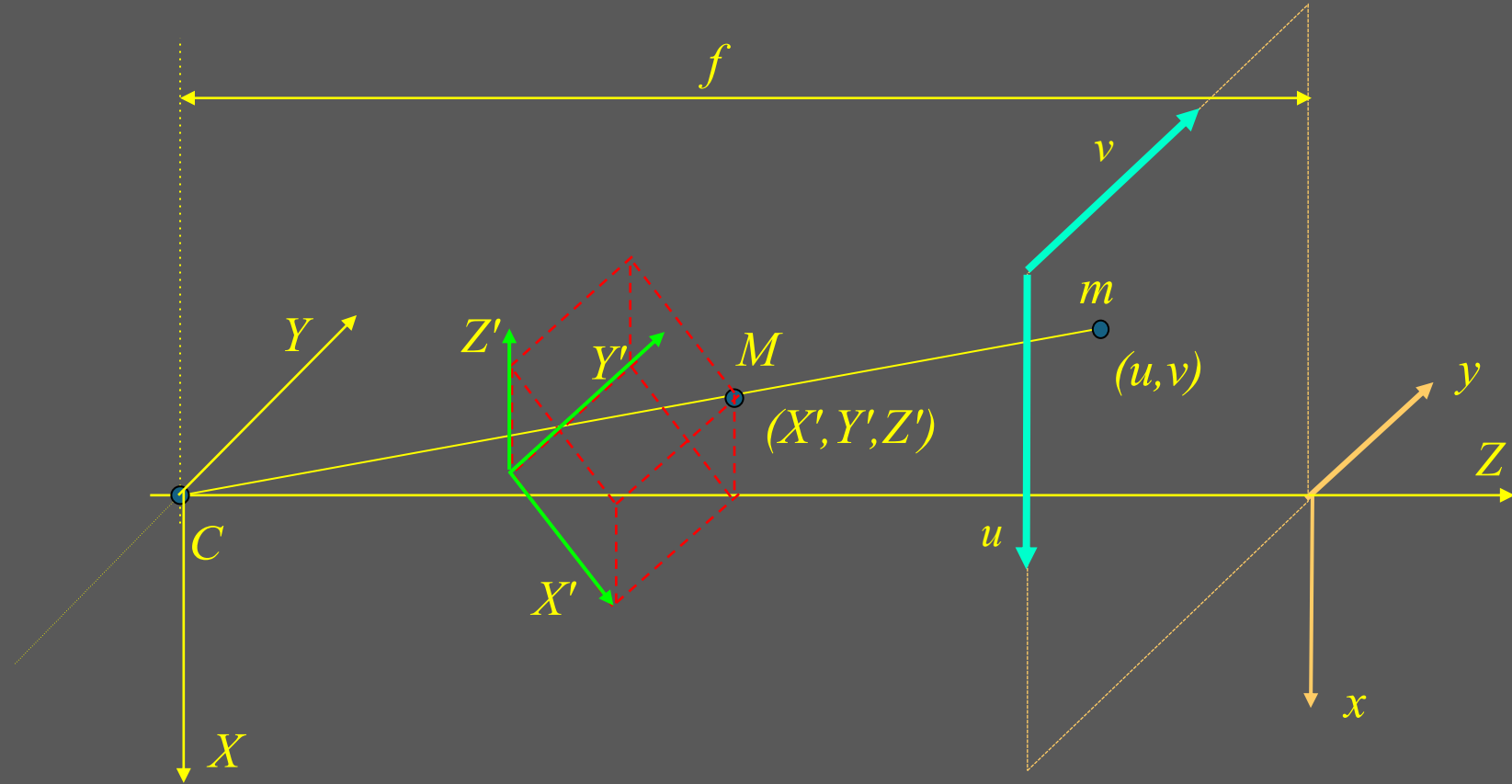
$$\mathbf{m} = [x \ y \ 1]^T$$

$$\mathbf{w} = [u \ v \ 1]^T$$

$$\mathbf{w} = \mathbf{K}\mathbf{m}$$

[Perspective Projection]

Now, m is given in a new coordinate system: (u, v)



$$\lambda \mathbf{w} = \mathbf{KPS}'\mathbf{M}'.$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f\alpha_x & fs & u_0 & 0 \\ 0 & f\alpha_y & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

II DoF:

- 5 intrinsec parameters $(\alpha, \beta, \gamma, u_0, v_0)$
- 6 extrinsec parameters $(\omega_X, \omega_Y, \omega_Z), (t_X, t_Y, t_Z)$